EPIDEMIOLOGY

Community Medicine (PSM) | 3rd Year

Hi Sammy!
Epidemiology is the
BACKBONE of PSM.
Let's nail it!

Epidemiology | PSM | 3rd Year | MBBS

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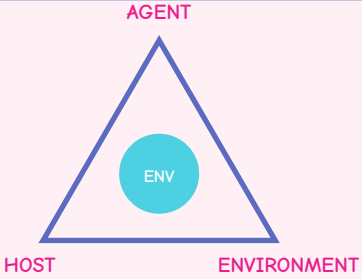
Pay attention here!
Study designs are asked
EVERY. SINGLE. YEAR.

DEFINITION

"The study of distribution and determinants of disease frequency in human populations." — John Last

Epidemiological Triad



Classification of Study Designs

Type	Subtype	Direction	Key Feature
Observational	Cross-sectional	Snapshot in time	Prevalence
	Case-control	Effect -> Cause	Odds Ratio
	Cohort	Cause -> Effect	Relative Risk
	Ecological	Group-level data	Correlations
Interventional	RCT	Researcher assigns	Gold standard
	Field trial	Healthy people	Vaccine trials
	Community trial	Whole community	Fluoridation

Memory Trick!
Case-control = looks BACKWARD (retrospective) = Odds Ratio
Cohort = looks FORWARD (prospective) = Relative Risk

THIS is the money page!
Formulas come EVERY
year. Learn them cold!

Measures of Disease Frequency

Measure	Formula	Key Point
Incidence Rate	New cases / Pop at risk x time	Measures RISK of getting disease
Prevalence	All cases / Total population	Snapshot — includes old + new
Attack Rate	Cases / Exposed in outbreak	Used in epidemics
Case Fatality	Deaths / Total cases x 100	Severity of disease
Mortality Rate	Deaths / Mid-year population	Risk of dying

★ REMEMBER:

Prevalence = Incidence x Duration of disease

(If incidence UP or duration UP, prevalence UP)

Measures of Association

Measure	Formula	Used In
Relative Risk	a/(a+b) / c/(c+d)	Cohort studies
Odds Ratio	ad / bc	Case-control studies
Attributable Risk	Ie - Iu	Excess risk due to exposure
PAR (Pop. Attr.)	Ip - Iu	Impact on population
NNT	1 / ARR	Treatment benefit

The Sacred 2x2 Table

	Disease+	Disease-
Exposed+	a	b
Exposed-	c	d

OR = ad/bc

RR = [a/(a+b)] / [c/(c+d)]

AR = a/(a+b) - c/(c+d)

★ Memory Trick!

Mnemonic: "OR = Odd Ratio = ad/bc = criss-cross"

"RR = Risk Ratio = used in cohoRt = foRwaRd looking"

Don't get confused!
Bias vs Confounding
is a classic trap Q!

Types of Bias

Bias Type	What Happens	Example / Prevention
Selection Bias	Non-random selection of subjects	Berkson's bias (hospital)
Information Bias	Errors in measuring exposure/outcome	Recall bias, observer bias
Recall Bias	Cases remember exposure better	Common in case-control
Observer Bias	Researcher influences measurement	Fix: blinding
Hawthorne Effect	Subjects change behavior if watched	Being observed = different
Lead Time Bias	Screening detects early, seems longer	Survival appears longer
Length Time Bias	Screening catches slow-growing disease	Overestimates benefit

Confounding

A confounding variable:

- 1. Associated with BOTH exposure AND outcome
- 2. NOT on the causal pathway between them
- 3. Unequally distributed between groups

Control: Randomization, Restriction, Matching, Stratification, Regression

Bradford Hill Criteria for Causation

1. **Strength** — Strong association = likely causal
2. **Consistency** — Reproduced in different settings
3. **Specificity** — One cause -> one effect
4. **Temporality** — Exposure BEFORE disease (ONLY essential)
5. **Dose-response** — More exposure -> more disease
6. **Plausibility** — Biologically makes sense
7. **Coherence** — Consistent with known facts
8. **Experiment** — Removal of cause -> less disease
9. **Analogy** — Similar agents -> similar effects

★ Memory Trick!
TEMPORALITY is the ONLY essential criterion! Exam favourite!

Clinical Application + Indian Context

Indian data is KEY!
Examiners love asking
about Indian programs.

Choosing the Right Study Design

Scenario	Best Design	Why
Rare disease	Case-control	Start from cases, look back
Rare exposure	Cohort	Follow exposed group forward
Causation proof	RCT	Gold standard, random allocation
Quick & cheap survey	Cross-sectional	One-time snapshot
Ethical constraints	Cohort (observational)	Can't randomize harmful exposure
Outbreak investigation	Case-control + descriptive	Fast, identifies source

Indian Epidemiological Context

- ICMR — Indian Council of Medical Research — apex body for health research
- NFHS — National Family Health Survey — demographic + health data (latest: NFHS-5)
- SRS — Sample Registration System — birth & death data by Registrar General
- IDSP — Integrated Disease Surveillance Programme — weekly outbreak alerts
- IHIP — Integrated Health Information Platform — replaced IDSP
- EIS — Epidemic Intelligence Service — rapid response teams

Levels of Prevention (Leavell + Clark)

Level	When	Examples
Primordial	Before risk factors	Health promotion, lifestyle changes
Primary	Before disease occurs	Vaccination, health education
Secondary	Early disease stage	Screening, early diagnosis
Tertiary	Established disease	Rehabilitation, prevent complications

★ Park's Textbook is THE reference for Indian PSM exams!

You've GOT this Sammy!
These Qs come every
year. Memorise them!

Most Repeated LAQs (10–12 marks)

- 1. Describe different types of epidemiological study designs with merits & demerits
- 2. What is a cohort study? Describe types, advantages, and disadvantages
- 3. Explain Bradford Hill criteria for causation with examples
- 4. Describe various types of bias in epidemiological studies. How to prevent?
- 5. Define and explain measures of disease frequency with formulas

Most Repeated SAQs (5–6 marks)

- 1. Differentiate case-control vs cohort study
- 2. Calculate OR and RR from 2x2 table
- 3. Define incidence and prevalence with relationship
- 4. What is confounding? How to control it?
- 5. Explain recall bias and observer bias
- 6. Short note on Hawthorne effect

LAQ Answer Blueprint

1	Define	Start with Park's definition (2-3 lines)
2	Classify	Draw classification tree / table
3	Compare	Merits vs demerits table (always!)
4	Diagram	2x2 table / triad / flowchart
5	Example	Real-world Indian example
6	Conclude	One-liner clinical significance

High Yield Checklist

MUST KNOW

- 2x2 table + OR/RR
- Study design types
- Incidence vs Prevalence
- Bradford Hill criteria
- Types of bias

HIGH YIELD

- Confounding control
- Levels of prevention
- Attack rate in outbreaks
- NNT calculation
- Case fatality rate

BONUS

- Web of causation
- Ecological fallacy
- DALY / QALY
- Sensitivity analysis

♥ **You're going to be an amazing doctor, Sammy!**

Epidemiology is just detective work for diseases. You've got this!